LD OF AGONISTS

being used for diabetes, obesity and heart health



SHUTTERSTOCK

Approved for what and when: For Type 2 diabetes in the US in 2022

Cost: $1,069 (Rs 88,600 approx.) for four injections. Not available in India

ZEPBOUND

Active ingredient: Tirzepatide (GIP and GLP-1 agonist)

Manufactured by: Eli Lilly

Form and frequency: Weekly injectable with a starting dose of 2.5 mg

Approved for what and when: For weight loss in the US in 2023

Cost: $1,060 (Rs 88,000 approx.) for four injections. Not available in India

VICTOZA

Active ingredient: Liraglutide (GLP-1 agonist)

Manufactured by: Novo Nordisk

Form and frequency: Daily injectable of 0.6 mg

Approved for what and when: For Type 2 diabetes in India in 2010

Cost: Rs 5,324 per month for an injection

APLEVANT

Active ingredient: Dulaglutide (GLP-1 agonist)

Manufactured by: Eli Lilly, sold by Lupin under collaboration

Form and frequency: Weekly injectable of 0.75 mg

Approved for what and when: For Type 2 diabetes in India in 2018

Cost: Rs 5,497 for a pre-filled injectable pen

BYETTA

Active ingredient: Exenatide (GLP-1 agonist)

Manufactured by: Eli Lilly

Form and frequency: Twice daily injection of .005 mg

Approved for what and when: Approved for Type 2 diabetes in India in 2007

Cost: Rs 6,957 for a 60-dose pen

Other GLP-1 drugs under research for diabetes and obesity

GLO034

GLP-1 agonist candidate by Sun Pharma currently being tested in non-obese and obese adults without diabetes in India

SEMAGLUTIDE INJECTION

The Central Drugs Standard Control Organisation has approved Dr. Reddy's Labs to carry out a bioequivalence study

GLP-1: Glucagon-like Peptide 1; GIP: Glucose-dependent Insulinotropic Polypeptide

of other diseases—from sleep apnea to fatty liver disease, and even Alzheimer's.

Which begs the question: Is Ozempic, the name that has come to represent all agonists, the answer to all our ills?

THE AGONISTS AND THE ECSTASY

Semaglutide falls under the class of drugs classified as glucagon-like peptide 1 or GLP-1 agonists, also known as GLP-1 receptor agonists, incretin mimetics, or GLP-1 analogs. Synthetic peptides, they mimic the actions of the naturally-occurring hormone. Novo Nordisk has been working on semaglutide since 2012, but GLP-1 medicines in themselves are not new. "They have been around for almost 20 years now," says Dr Ambrish Mithal, chairman and head of the endocrinology and diabetes division at Max Healthcare Hospitals. In fact, the FDA approved the first GLP-1 agonist, exenatide (Byetta), in 2005. Other GLP-1 agonists include the injectable dulaglutide (Trulicity), liraglutide (Victoza) and lixisenatide (Adlyxin). With 391 patents and involvement in 262 trials, semaglutide has outshone them all because of its encouraging results, significant weight reduction and ease of use. In May 2022, however, US pharma giant Eli Lilly received FDA approval for its drug tirzepatide, which mimics two endocrine hormones responsible for the release of insulin—GLP-1 and GIP (glucose-dependent insulinotropic polypeptide). For this reason, tirzepatide has proven to be more effective in weight loss and sugar control. It is marketed as Mounjaro for diabetes and Zepbound for weight loss.

But how do GLP-1 agonists work (see *Semaglutide FAQs*)? Semaglutide binds to the GLP-1 receptors found in the pancreas and in some neurons in the brain and activates them once food is ingested. This, in turn, persuades the beta cells in the pancreas to produce insulin, which transports glucose into our cells for energy. Through complex endocrine mechanisms, semaglutide also reduces the release of another